LAB 1: Exploring website terminologies and architecture

Objectives To engage students in a technical exploration of websites by analyzing URLs, browser behavior, web content and website architecture

Theoretical description

1. What is a Website?

- **Definition:** A website is a collection of interlinked web pages, typically served from a single domain name and hosted on a web server.
- **Components:**
  - **Web Pages:** Documents that are formatted using markup languages like HTML (focus here is on understanding, not coding).
  - **Media Content:** Images, videos, and other multimedia elements.
  - **Navigation:** How users move from one page to another within the website.
  - **User Interface (UI):** How the website looks and how users interact with it.

2 Website Architecture

- **Client-Server Model:**
  - **Client:** The web browser that requests web pages.
  - **Server:** The web server that responds to these requests by serving the web pages.
- **HTTP/HTTPS Protocols:** The rules governing the exchange of information between the client and server.
- **Domain Names and DNS:**
  - **Domain Name:** The human-readable address of a website (e.g.www.example.com).
  - **DNS (Domain Name System):** Translates domain names into IP addresses so browsers can load internet resources.

3 Key Concepts in Web Engineering

- **Web Accessibility:** Ensuring that websites can be used by people with disabilities.
- **User Experience (UX):** Designing websites to be easy and intuitive for users.
- **Scalability:** The ability of a website to handle increasing amounts of traffic.
- **Security:** Protecting websites from threats like hacking and data breaches.

4. Web Engineering Principles

Usability: Ensuring that a website is e+asy to use and navigate.
Performance: The speed at which a website loads and operates.
Scalability: The ability of a website to grow and handle more traffic.
Security: Protecting the website from unauthorized access and data breaches.

TASKS

1. Analyzing URL

- Visit several websites (e.g., https://www.example.com, https://www.wikipedia.org).
- For each URL, break it down into its components: protocol (e.g., https), domain (e.g., www.wikipedia.org), and path (e.g., /wiki/Main_Page).
- Modify the URL paths by adding, removing, or changing parts of the URL. For example, try removing the last part of the URL and observe the browser's response.

Questions:

- What happens when you remove or modify the path in the URL?
- How does the browser respond when you enter an incorrect path? (e.g., 404 error)
- What do you notice about URLs that use http versus https?

2. Browser Behavior and Rendering Analysis

- Open the same website i.e in part 1 (e.g., https://www.bbc.com) in different browsers (Chrome, Firefox, Edge).
- Observe the loading time, layout, and rendering of content.
- Inspect elements on the page using the browser's developer tools (right-click -> Inspect). Focus on differences in how elements are rendered.

Questions:

- Are there any noticeable differences in how the website is displayed across different browsers?
- What differences do you observe in loading times? What might cause these differences?
- How do browser developer tools help you understand the structure and rendering of a webpage?

3. Analyzing web content types and file formats

- Download various types of content from websites (e.g., images, PDFs, videos).
- Identify the file formats (e.g., .jpg, .pdf, .mp4) and note the size of each file.
- Use the browser's "Save As" feature to save a webpage and observe the types of files it generates (e.g., HTML, CSS, JS).

Questions:

- How do different file formats contribute to the overall content of a website?
- What is the significance of file size in terms of web performance?
- What files are saved when you download a webpage, and what role does each file play in rendering the page?

4. Exploring Website Structure

1. Select a popular website (e.g., Wikipedia, Amazon) to explore.
2. Observe how the website is organized (home page, subpages, main sections and navigation path).
3. Identify different types of pages (e.g., landing pages, home pages, content pages, forms).
4. Write a brief summary of the website’s structure and how the navigation is organized.
5. Create a simple flowchart or diagram showing the structure and hierarchy of the website’s pages
6. Express user experience of any one website selected above